

# Generation Interconnection System Impact Study Report

## For Queue Project Q22-2130

Sycan 138 kV  
October 19, 2023

### **1. Introduction**

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2130. The Interconnection Customer proposes a 140.7 MW solar photovoltaic facility interconnecting at the Sycan 138 kV substation in the Virginia area. The request was received on July 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 655 higher-queued active requests and their assigned network upgrades.

### **2. Summary of Results**

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$4,887,000.

### **3. Network Impacts**

#### **3.1 Generator Deliverability and Single Contingency Analysis**

1. Facility: Hartman Crk 138/1kV Transformer Bank  
Contingency: Lowmoor to Hinton Ap 345 KV  
Loading: 126.2% (Rating: 490 MVA)  
DFAX: 13.4%  
Pre-project loading: 122.3%
2. Facility: Hartman Crk 345 kV/1 kV Transformer  
Contingency: Lowmoor to Hinton Ap 345 KV L  
Loading: 121.5% (Rating: 509 MVA)  
DFAX: 13.4%  
Pre-project loading: 117.7%
3. Facility: Killdeer to Hyak 138 KV Line  
Contingency: Stock Island to Kentucky Dam 138 KV L  
Loading: 121.3% (Rating: 179 MVA)  
DFAX: 52.7%  
Pre-project loading: 79.8%
4. Facility: Sycan to Killdeer 138 KV Line  
Contingency: Kentucky Dam Substation to Stock Island 138 KV L  
Loading: 118.8% (Rating: 188 MVA)  
DFAX: 52.7%  
Pre-project loading: 79.4%
5. Facility: Piute to Bluefield Avenue 138 KV line  
Contingency: No Troy to Lost Nation 138 KV  
Loading: 117.1% (Rating: 198 MVA)  
DFAX: 13.1%  
Pre-project loading: 107.9%
6. Facility: Nassau to Litchfield 138 KV Line  
Contingency: No Troy to Lost Nation 138 KV L  
Loading: 107.6% (Rating: 231 MVA)

DFAX: 13.1%  
Pre-project loading: 99.6%

7. Facility: Springbank to Litchfield 138 KV  
Contingency: No Troy to Lost Nation 138 KV L  
Loading: 106.2% (Rating: 257 MVA)  
DFAX: 13.1%  
Pre-project loading: 99.0%

8. Facility: Friendship to Hyak Substation 138 KV  
Contingency: Orovada to Tolson Substation 138 KV  
Loading: 106.1% (Rating: 292 MVA)  
DFAX: 36.8%  
Pre-project loading: 88.3%

9. Facility: Hiton Ap to Manatee 345 KV  
Contingency: Longview2 Substation to Hartman Crk 345 KV L  
Loading: 101.2% (Rating: 1,150 MVA)  
DFAX: 23.0%  
Pre-project loading: 98.3%

### **3.2 Multiple Facility Contingency Analysis**

No additional overloads were identified for multiple facility contingencies.

### **3.3 Short Circuit and Stability**

Short circuit and stability analyses identified no additional network upgrades.

## **4. Required Network Upgrades and Cost Estimates**

| Facility                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------|---------------------------------------------|---------------|-----------------------|
| Hrtman Creek 138 kV/1 kV XFMR           | Replace transformer with higher-rated unit  | \$3,800,000   | 2% (\$87,000)         |
| Hartman Creek 345-1 kV Transformer Bank | Replace transformer with higher-rated unit  | \$13,974,000  | 2% (\$320,000)        |
| Killdeer to Hyak 138 KV Line            | Rebuild line with higher-capacity conductor | \$6,489,000   | 18% (\$1,168,000)     |
| Killdeer to Sycan 138 KV                | Rebuild line with higher-capacity conductor | \$6,053,000   | 18% (\$1,089,000)     |
| Piute to Bluefield Avenue 138 KV line   | Rebuild line with higher-capacity conductor | \$17,900,000  | 6% (\$994,000)        |
| Nassau to Litchfield 138 KV Line        | Reconductor line                            | \$3,107,000   | 6% (\$172,000)        |
| Springbank to Litchfield 138 KV         | Reconductor line                            | \$3,670,000   | 6% (\$204,000)        |
| Friendship to Hyak Substation 138 KV    | Reconductor line                            | \$1,190,000   | 10% (\$121,000)       |
| Hiton Ap to Manatee 345 KV              | Reconductor line                            | \$28,682,000  | 3% (\$732,000)        |
|                                         | Total allocated to project                  |               | \$4,887,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.